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Luke Conaboy

Computational astrophysicist researching the first galaxies and the intergalactic medium using large-scale cosmological radiation-hydrodynamical simulations and machine learning.

Employment

2023–present Research Fellow in Computational Astrophysics, *University of Nottingham*, Nottingham, UK
Advisor: Prof James Bolton

2022–2023 Research Associate, *University of Nottingham*, Nottingham, UK
Advisor: Dr Emma Chapman

Education

2018–2022 PhD Astronomy, *University of Sussex*, Brighton, UK
Supervisor: Prof Ilian Iliev
Thesis: Simulations of structure formation and feedback at high redshift

2014–2018 MSci Physics with Astronomy, *University of Nottingham*, Nottingham, UK
Degree classification: First
Thesis: Investigating the response in African climate anomalies to the El Niño–Southern Oscillation

Computing skills

I am a proficient user of various programming languages and numerical modelling codes, those which I have used extensively are listed below. I am also a developer and contributor to open source projects, selected projects are listed below.

Languages Python (`tensorflow`, `pytorch`, `numpy`, `scipy`, `matplotlib`, `mpi4py`, `yt`), Fortran, bash/sh, MPI, C/C++, Matlab, `git`

Codes `ramses`, `music`, `camb`, `gadget`, `aton`

Other Linux, High performance computing, LaTeX, GitHub

Projects `drft`, Python/MPI/Fortran code used in Conaboy et al. (2023) to modify simulation initial conditions to include the effect of supersonic relative baryon–dark matter velocities.
Repository: `https://github.com/lconaboy/drft`

`r2g`, Version of open source MPI/Fortran simulation format conversion code, modified to analyse the petabyte-scale cosmological simulation Cosmic Dawn III.
Repository: `https://github.com/lconaboy/r2g`

Publications

I am an author on eight peer-reviewed articles (plus three in review, and two in preparation), three of which I am the lead author (118 citations, h-index $= 6$). Publications marked with an asterisk indicate work led by students where I played a major role in supervision or collaboration.

First author

2026 *Impact of source and survey modelling on the galaxy–Ly α transmission correlation*, **Conaboy**, Bolton, Keating, Haehnelt, Kulkarni, Puchwein, in prep.

2025 *The connection between high-redshift galaxies and Lyman α transmission in the Sherwood-Relics simulations of patchy reionisation*, **Conaboy**, Bolton, Keating, Haehnelt, Kulkarni, Puchwein, MNRAS 539(3)

2023 *Relative baryon-dark matter velocities in cosmological zoom simulations*, **Conaboy**, Iliev, Fialkov, Dixon, Sullivan, MNRAS 525(4)

Coauthor

2026 *Double-Peaked Ly Emission during Reionization Requires Nearby Voids and a Favorable Local Ionizing Background*, Park, Smith, Jung, Yajima, Ocvirk, Lewis, Conaboy, Shapiro, Iliev, Ahn, Lee, Sorce, Dubois, arXiv e-prints

***** *Temperature measurements from quasar spectra with machine learning*, Sawyer, Bolton, **Conaboy**, and Becker, in prep.

2025 *Dust-UV offsets in high-redshift galaxies in the Cosmic Dawn III simulation*, Ocvirk, Lewis, **Conaboy**, Dubois, Bethermin, Sorce, Aubert, Shapiro, Dawoodbhoy, Lee, Teyssier, Yepes, Gottlöber, Iliev, Ahn, and Park, A&A 703

Square Kilometre Array Science Data Challenge 3a: Foreground removal for an EoR experiment, SKA Data Challenge participants, **Conaboy**, MNRAS 543(2)

***** *Reionization in HESTIA: Studying reionization in the LG through zoom simulations*, Attard, **Conaboy**, Libeskind, Pillipenko, Dixon, Iliev, arXiv e-prints

The Thermal Sunyaev-Zel'dovich Effect from the Epoch of Reionization, Iliev, Hosein, Chluba, **Conaboy**, Attard, Mondal, Ahn, Gottlöber, Lewis, Ocvirk, Park, Shapiro, Sorce, and Yepes, MNRAS 540(2)

How probable is the Lyman-α damping wing in the spectrum of the redshift $z = 5.9896$ quasar ULAS J0148+0600?, Sawyer, Bolton, Becker, **Conaboy**, Haehnelt, Keating, Kulkarni, Puchwein, MNRAS 539(3)

2024 *Reproducibility of machine learning analyses of 21 cm reionization maps*, Sooknunan, Chapman, **Conaboy**, Mortlock, and Pritchard, arXiv e-prints

***** *The Lyman-limit photon mean free path at the end of late reionization in the Sherwood–Relics simulations*, Feron, **Conaboy**, Bolton, Chapman, Haehnelt, Keating, Kulkarni, Puchwein, MNRAS 532(2)

2022 *The short ionizing photon mean free path at z=6 in Cosmic Dawn III, a new fully-coupled radiation-hydrodynamical simulation of the Epoch of Reionization*, Lewis, Ocvirk, Sorce, Aubert, **Conaboy**, Shapiro, Dawoodbhoy, Teyssier, Yepes, Gottlöber, Ahn, Iliev, Thélie, MNRAS 516(3)

Grants

I have been involved in multiple successful applications for computing time, both at national and international facilities. The total monetary value of these successful applications is over £1M in compute time, of which £660k is as Co-PI. The approximate monetary value of each compute time award is estimated using the core-hour rate offered by the University of Nottingham of £0.015 per core-hour (Oct 2025).

Co-PI

2025 JUWELS (Jülich, Germany) computing time (PI: Dr Noam Libeskind, PC: Prof Ilian Iliev) **14.2M core-h**, *£213k*

2024 JUWELS (Jülich, Germany) computing time (PI: Dr Noam Libeskind, PC: Prof Ilian Iliev) **8.8M core-h**, *£132k*

2023 JUWELS (Jülich, Germany) computing time (PI: Dr Noam Libeskind, PC: Prof Ilian Iliev) **10.0M core-h**, *£150k*

2022 JUWELS (Jülich, Germany) computing time (PI: Dr Noam Libeskind, PC: Prof Ilian Iliev) **11.0M core-h**, *£165k*

Contributor

2025 EuroHPC, LUMI-C (Kajani, Finland) computing time (PI: Dr Laura Keating) **15.1M core-h**, *£227k*

2024 DiRAC (Cambridge, Edinburgh, and Durham, UK) computing time (PI: Prof James Bolton) **12.7M core-h**, *£190k*

2021 PRACE, Beskow/Dardel KTH (Stockholm, Sweden) computing time (PI: Prof Ilian Iliev) **12.0M core-h**, *£180k*

Conference proceedings

2022 *The Reionisation of the Local Universe in the Hestia Suite*, *John von Neumann Institute for Computing Symposium*, **Conaboy**, Iliev, Libeskind

Supervision

Assistant supervisor

2022-present Fiona Sawyer, PhD, University of Nottingham

2022-2026 Jennifer Feron, PhD, University of Nottingham

Selected talks

Apr 2026 Charting Cosmic Dawn in Copenhagen, Helsingør

Mar 2026 Astronomy seminar, online, **invited**

Jan 2026 E.A. Milne Centre for Astrophysics 10th Anniversay, University of Hull

Oct 2025 CCS Colloquium, CCS, Tsukuba, **invited**

Oct 2025 KMM Discussion Meeting, NAOJ, Tokyo, **invited**

Sep 2025 Kaba Kada EoR Meeting, Port Douglas

Apr 2025 Cake Talk, DAWN, Copenhagen
Feb 2024 Astronomy Colloquium, ICC, Durham, **invited**
Nov 2023 TIFR State of the Universe, TIFR, Mumbai (online)
Nov 2023 Cambridge Galaxies Discussion Group, KICC, Cambridge, **invited**
Jun 2023 CLUES Meeting 2023, LMU CAS, Munich
Jul 2022 CLUES Meeting 2022, UAM, Madrid
Sep 2021 RAMSES User Meeting 2021, online
Dec 2020 PhD in a Pandemic 2020, online
Nov 2020 CLUES Mid-term Meeting, online
Oct 2020 Third Global 21cm Meeting, online
Oct 2020 RAS Specialist Meeting, London, UK

Awards and press releases

2022 Physics Finalist, STEM for Britain
Selected to present a poster in the Houses of Parliament to members of both Houses. Attendance covered in local news media, including BBC radio.

Conference, workshop, and seminar organisation

2025 EAS 2025 SS22 – Modelling the first billion years, *SOC*, Cork, Ireland
Selecting talks

2022–present University of Nottingham Astronomy Seminar Series, *Organiser*, Nottingham, UK
Inviting and hosting guest speakers

2020 SAZERAC 2020, *LOC*, online
Moderating discussion

Teaching

2020 Foundation Mathematics B, *University of Sussex*
Assisting with workshops, marking.

2019 Scientific Computing, *University of Sussex*
Assisting with workshops, marking.

2019 Financial Computing with MATLAB, *University of Sussex*
Assisting with workshops, marking.

Languages

native English
beginner Danish, French (GCSE)

References

Prof James Bolton, *School of Physics and Astronomy, The University of Nottingham, University Park, Nottingham, NG7 2RD, UK*, `james.bolton@nottingham.ac.uk`

Prof Ilian Iliev, *Astronomy Centre, Department of Physics & Astronomy, University of Sussex, Brighton, BN1 9QH, UK*, i.t.iliev@sussex.ac.uk

Dr Pierre Ocvirk, *Observatoire Astronomique de Strasbourg - UMR 7550 11 rue de l'Université 67000 Strasbourg France*, pierre.ocvirk@astro.unistra.fr